

Estimated Annual Agricultural Pesticide Use Malheur County, Oregon 2014-2018

24 July 2026

Summary

Malheur County ranks **#3 of Oregon's 36 counties** by total estimated agricultural pesticide mass applied (EPest-high method, summed 2014-2018), making it one of OR's five highest pesticide-use counties. Malheur County, in Oregon's Treasure Valley, is a leading onion-growing region, also producing potatoes and sugar beets under irrigation from the Owyhee and Snake River systems.

Table 2 lists the rank and best available estimates of mass applied (kg) for the top 80 agricultural pesticides in **Malheur County, Oregon** during 2018, as reported by [USGS Estimated Annual Agricultural Pesticide Use](#) data. The table also includes rank by 3-year average (2016-2018) and 5-year average (2014-2018) estimates of mass applied, and each compound's chemical class (abbreviated; see the key in Table 1 and the full classification, with a separate functional class column, in `data/pesticide_classification.csv`, shared with every chapter in this repository). Rankings and estimates are based on the *EPest-high method* described by [Thelin and Stone \(2013\)](#) and [Baker and Stone \(2015\)](#):

“EPest-low and EPest-high, are used to estimate a range of pesticide use. Both EPest-low and EPest-high methods incorporate proprietary surveyed rates for Crop Reporting Districts (CRDs), but EPest-low and EPest-high estimates differ in how they treat situations when a CRD was surveyed and pesticide use was not reported for a particular crop present in the CRD. In these situations, EPest-low assumes zero use in the CRD for that pesticide-by-crop combination. EPest-high, however, treats the unreported use for that pesticide-by-crop combination in the CRD as missing data. In this case, pesticide-by-crop use rates from neighboring CRDs or CRDs within the same region are used to estimate the pesticide-by-crop EPest-high rate for the CRD.” -[Pesticide National Synthesis Project webpage](#)

USGS lists several caveats of these data, in particular:

- Reliability decreases with scale (e.g. “detailed interpretation of where and how much use occurs within a county is not appropriate”).
- EPest-low estimates are more likely to reflect state-based restrictions on pesticide use than EPest-high estimates.
- A blank cell in the source file (no reported/estimated use for that compound-year-county) is treated as zero in this chapter.
- **2019 is excluded from this ranking.** The 2019 county-level file available at the time of writing covers only 69 compounds nationwide, versus roughly 400 compounds in every other year from 1992-2018 — a partial/preliminary release, not a real drop in the number of pesticides in use. Anchoring the ranking on it would silently omit hundreds of real compounds. This chapter uses 2018 (the most recent complete year) as the 1-year snapshot instead; re-run with 2019 -included windows once a complete 2019 release is available.

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)

Code	Chemical class
CPX	Chlorophenoxy compound
DIP	Dipyridyl compound
DTC	Dithiocarbamates
INO	Inorganic compounds
MIC	Microbial
CB	N-methyl carbamates (AChE inhibitor)
OGM	Organo-metallic compound
OC	Organochlorine compound
OP	Organophosphorous compound

Table 1: Chemical class abbreviation key (used in the Class column of Table 2)
(continued)

Code	Chemical class
OTH	Other
PYR	Pyrethrins/Pyrethroids
TC	Thiocarbamates
TRZ	Triazines

Table 2: Top 80 EPest-high Estimates, Malheur County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
1	BACILLUS AMYLOLIQUEFACIEN	MIC	464275	159527	159527
2	CHLOROPICRIN	OC	383043	156179	108250
3	METAM	DTC	296851	474428	372053
4	GLYPHOSATE	OTH	54825	37175	34543
5	DICHLOROPROPENE	OC	39603	41840	127575
6	BURKHOLDERIA SPP	MIC	31926	32923	32923
7	METAM POTASSIUM	DTC	14577	152884	118572
8	METHOMYL	CB	14163	14830	14428
9	PENDIMETHALIN	OTH	11321	11147	10655
10	TRIFLURALIN	OTH	9988	8606	12235
11	SULFURIC ACID	INO	7628	18102	20198
12	ATRAZINE	TRZ	6178	8383	6736
13	MANCOZEB	DTC	4998	4426	3703
14	2,4-D	CPX	4921	6923	5411
15	BROMOXYNIL	OTH	4607	5184	4860
16	ETHOFUMESATE	OTH	4178	5837	4597
17	METOLACHLOR & METOLACHLOR-S	OTH	3946	4705	5629
18	ETHOPROPHOS	OP	3921	2168	1307
19	EPTC	TC	3663	4500	5387
20	MALATHION	OP	3448	4671	3854
21	METOLACHLOR-S	OTH	3382	3416	3900
22	DIMETHENAMID & DIMETHENAMID-P	OTH	3169	4662	3443
23	DIMETHENAMID-P	OTH	3169	2373	2069
24	SULFUR	INO	2787	3874	3444
25	MCPA	CPX	2474	4849	4266
26	CHLOROTHALONIL	OTH	2418	8314	6984
27	CLETHODIM	OTH	2021	2099	1825
28	BOSCALID	OTH	1979	1429	1378
29	CHLORPYRIFOS	OP	1910	2311	2878
30	FLUFENACET	OTH	1814	1315	1325
31	THIOPHANATE-METHYL	OTH	1733	2441	1634
32	METRIBUZIN	TRZ	1477	955	1145
33	OXYFLUORFEN	OTH	1348	1522	1268
34	ETHALFLURALIN	OTH	1341	2160	2059
35	PARAQUAT	DIP	1302	991	998
36	PYRACLOSTROBIN	OTH	1285	1509	1380
37	COPPER HYDROXIDE	INO	1127	2861	2835
38	DIURON	OTH	996	2372	3239
39	DICAMBA	OTH	904	2030	1714
40	SPIROTETRAMAT	OTH	890	813	730
41	PENTHIOPYRAD	OTH	787	798	590
42	IMIDACLOPRID	OTH	770	673	731
43	IPRODIONE	OTH	738	2167	2291
44	OXAMYL	CB	722	462	3209
45	SPINETORAM	OTH	702	582	508
46	ACETOCHLOR	OTH	695	1439	1671

Table 2: Top 80 EPest-high Estimates, Malheur County, OR, 1-Yr (2018), 3-Yr Avg, 5-Yr Avg, with Chemical Class (abbreviated; see the classification key table above) (*continued*)

Rank	Compound	Class	EPest-high 1yr	EPest-high 3yr avg	EPest-high 5yr avg
47	COPPER OXYCHLORIDE	INO	672	349	843
48	TERBACIL	OTH	670	670	466
49	DIMETHOATE	OP	636	840	1098
50	AZOXYSTROBIN	OTH	591	636	531
51	MEFENOXAM	OTH	570	891	832
52	METOLACHLOR	OTH	564	1289	1730
53	PROPICONAZOLE	OTH	542	773	750
54	CLOPYRALID	OTH	520	545	647
55	BENTAZONE	OTH	461	311	483
56	DIMETHOMORPH	OTH	394	645	653
57	FENAMIDONE	OTH	348	348	348
58	MESOTRIONE	OTH	328	282	264
59	DIQUAT	DIP	316	250	242
60	CYHALOTHRIN-LAMBDA	PYR	301	229	236
61	HEXYTHIAZOX	OTH	282	737	676
62	AMETOCTRADIN	OTH	268	475	475
63	ALPHA CYPERMETHRIN	PYR	257	136	136
64	SPINOSYN	OTH	246	98	394
65	DICLORAN	OTH	246	354	642
66	PYMETROZINE	OTH	217	122	98
67	FLUROXYPYR	OTH	209	1010	706
68	FLUMIOXAZIN	OTH	209	77	55
69	BIFENTHRIN	PYR	204	194	197
70	FLUOPYRAM	OTH	194	102	78
71	DIFLUFENZOPYR	OTH	182	88	147
72	FLUXAPYROXAD	OTH	173	384	251
73	SULFENTRAZONE	OTH	172	85	79
74	PYRIMETHANIL	OTH	169	233	283
75	PINOXADEN	OTH	167	73	70
76	DIFENOCONAZOLE	OTH	164	155	137
77	IMAZETHAPYR	OTH	164	172	137
78	TETRACONAZOLE	OTH	163	106	72
79	ABAMECTIN	OTH	162	111	77
80	HEXAZINONE	TRZ	138	920	729

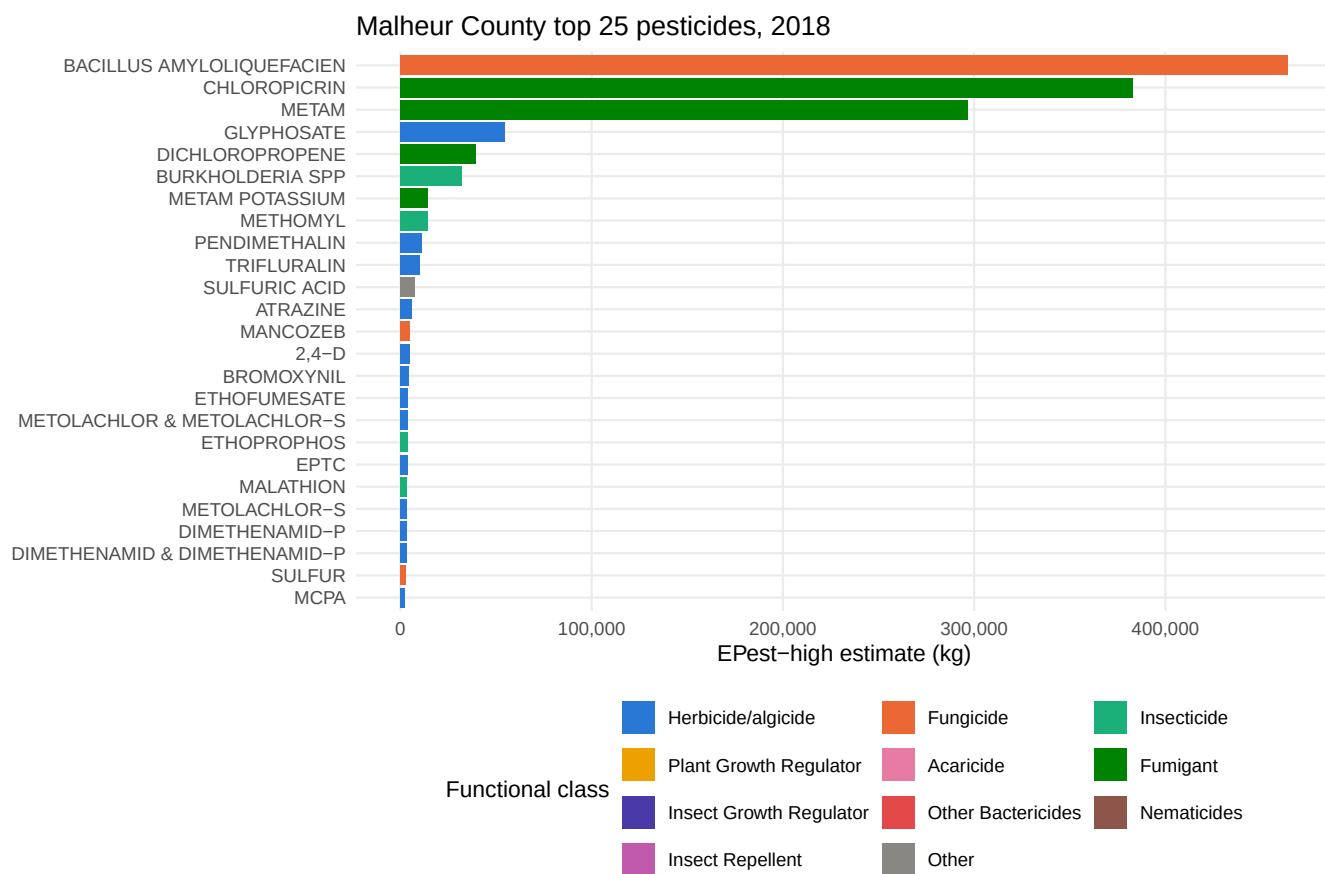


Figure 1: Top 25 compounds in Malheur County by E Pest-high mass applied, most recent year, colored by pesticide functional class.

Appendices

Table 3: Top 50 Malheur County Estimates, Range (EPest-low and EPest-high), 1-Yr, 2018

RANK_1YR	COMPOUND	EPEST_LOW_KG_1YR	EPEST_HIGH_KG_1YR
1	BACILLUS AMYLOLIQUEFACIEN	0	464275
2	CHLOROPICRIN	375319	383043
3	METAM	263734	296851
4	GLYPHOSATE	54225	54825
5	DICHLOROPROPENE	39603	39603
6	BURKHOLDERIA SPP	0	31926
7	METAM POTASSIUM	0	14577
8	METHOMYL	14099	14163
9	PENDIMETHALIN	11321	11321
10	TRIFLURALIN	78	9988
11	SULFURIC ACID	0	7628
12	ATRAZINE	6178	6178
13	MANCOZEB	4998	4998
14	2,4-D	4809	4921
15	BROMOXYNIL	4455	4607
16	ETHOFUMESATE	4178	4178
17	METOLACHLOR & METOLACHLOR-S	3574	3946
18	ETHOPROPHOS	0	3921
19	EPTC	2698	3663
20	MALATHION	3448	3448
21	METOLACHLOR-S	3010	3382
22	DIMETHENAMID & DIMETHENAMID-P	3169	3169
23	DIMETHENAMID-P	3169	3169
24	SULFUR	2787	2787
25	MCPA	1461	2474
26	CHLOROTHALONIL	2418	2418
27	CLETHODIM	2021	2021
28	BOSCALID	1965	1979
29	CHLORPYRIFOS	1795	1910
30	FLUFENACET	0	1814
31	THIOPHANATE-METHYL	352	1733
32	METRIBUZIN	317	1477
33	OXYFLUORFEN	1348	1348
34	ETHALFLURALIN	1341	1341
35	PARAQUAT	273	1302
36	PYRACLOSTROBIN	1060	1285
37	COPPER HYDROXIDE	926	1127
38	DIURON	330	996
39	DICAMBA	695	904
40	SPIROTETRAMAT	879	890
41	PENTHIOPYRAD	762	787
42	IMIDACLOPRID	730	770
43	IPRODIONE	0	738
44	OXAMYL	248	722
45	SPINETORAM	702	702
46	ACETOCHLOR	695	695
47	COPPER OXYCHLORIDE	471	672
48	TERBACIL	670	670
49	DIMETHOATE	378	636
50	AZOXYSTROBIN	538	591

Table 4: Top 50 Malheur County Estimates, Range (EPest-low and EPest-high), 3-Yr Avg, 2016-2018

RANK_3YR	COMPOUND	EPEST_LOW_KG_3YR_AVG	EPEST_HIGH_KG_3YR_AVG
1	METAM	463388	474428
2	BACILLUS AMYLOLIQUEFACIEN	0	159527
3	CHLOROPICRIN	126966	156179
4	METAM POTASSIUM	0	152884
5	DICHLOROPROPENE	35809	41840
6	GLYPHOSATE	35573	37175
7	BURKHOLDERIA SPP	0	32923
8	SULFURIC ACID	0	18102
9	METHOMYL	14781	14830
10	PENDIMETHALIN	11084	11147
11	TRIFLURALIN	223	8606
12	ATRAZINE	8339	8383
13	CHLOROTHALONIL	1136	8314
14	2,4-D	6725	6923
15	DIMETHENAMID	6868	6868
16	PHOSPHORIC ACID	0	6853
17	ETHOFUMESATE	5837	5837
18	BROMOXYNIL	5092	5184
19	MCPA	2552	4849
20	METOLACHLOR & METOLACHLOR-S	4100	4705
21	MALATHION	4062	4671
22	DIMETHENAMID & DIMETHENAMID-P	4356	4662
23	EPTC	2472	4500
24	MANCOZEB	4426	4426
25	SULFUR	3874	3874
26	METOLACHLOR-S	2811	3416
27	COPPER HYDROXIDE	2784	2861
28	PROPARGITE	2800	2800
29	THIOPHANATE-METHYL	357	2441
30	DIMETHENAMID-P	2067	2373
31	DIURON	575	2372
32	CHLORPYRIFOS	2237	2311
33	ETHOPROPHOS	220	2168
34	IPRODIONE	0	2167
35	ETHALFLURALIN	1292	2160
36	CLETHODIM	2006	2099
37	DICAMBA	1774	2030
38	OXYFLUORFEN	1521	1522
39	PYRACLOSTROBIN	1422	1509
40	ACETOCHLOR	2177	1439
41	BOSCALID	1446	1429
42	CHROMOBACTERIUM SUBTUGAE	0	1325
43	FLUFENACET	0	1315
44	METOLACHLOR	1289	1289
45	FLUROXYPYR	920	1010
46	PARAQUAT	406	991
47	METRIBUZIN	473	955
48	NALED	0	923
49	HEXAZINONE	378	920
50	MEFENOXAM	107	891

Table 5: Top 50 Malheur County Estimates, Range (EPest-low and EPest-high), 5-Yr Avg, 2014-2018

RANK_5YR	COMPOUND	EPEST_LOW_KG_5YR_AVG	EPEST_HIGH_KG_5YR_AVG
1	METAM	359970	372053
2	BACILLUS AMYLOLIQUEFACIEN	0	159527
3	DICHLOROPROPENE	123957	127575
4	METAM POTASSIUM	17229	118572
5	CHLOROPICRIN	90722	108250
6	GLYPHOSATE	32756	34543
7	BURKHOLDERIA SPP	0	32923
8	SULFURIC ACID	0	20198
9	METHOMYL	14357	14428
10	TRIFLURALIN	158	12235
11	PENDIMETHALIN	10511	10655
12	DCPA	0	10508
13	CHLOROTHALONIL	969	6984
14	DIMETHENAMID	6868	6868
15	ATRAZINE	7051	6736
16	DIAZINON	9	6601
17	METOLACHLOR & METOLACHLOR-S	5087	5629
18	2,4-D	5206	5411
19	EPTC	3209	5387
20	METHYL BROMIDE	5070	5070
21	BROMOXYNIL	4706	4860
22	ETHOFUMESATE	4597	4597
23	PHOSPHORIC ACID	14	4420
24	MCPA	2360	4266
25	METOLACHLOR-S	3359	3900
26	MALATHION	2699	3854
27	MANCOZEB	3703	3703
28	SULFUR	3444	3444
29	DIMETHENAMID & DIMETHENAMID-P	3035	3443
30	DIURON	421	3239
31	OXAMYL	2986	3209
32	CHLORPYRIFOS	2531	2878
33	COPPER HYDROXIDE	2782	2835
34	PROPARGITE	2682	2581
35	IPRODIONE	3	2291
36	DIMETHENAMID-P	1661	2069
37	ETHALFLURALIN	1538	2059
38	CLETHODIM	1694	1825
39	METOLACHLOR	1728	1730
40	DICAMBA	1447	1714
41	ACETOCHLOR	1796	1671
42	THIOPHANATE-METHYL	237	1634
43	PYRACLOSTROBIN	1290	1380
44	BOSCALID	1379	1378
45	FLUFENACET	0	1325
46	ETHOPROPHOS	138	1307
47	OXYFLUORFEN	1267	1268
48	TERBUFOS	1232	1232
49	METRIBUZIN	571	1145
50	DIMETHOATE	308	1098